



Probability Theory

Books:

- a. Wackerly, D. D., Mendenhall, W., & Scheaffer, R. L. (2008). Mathematical statistics with applications (7th ed.). Cengage Learning.
- b. Casella, G., & Berger, R. L. (2023). Statistical inference (2nd ed.). Routledge.
- c. Hayter, A., (2012). Probability & Statistics for Scientists and Engineers (4th Ed.). Brooks/Cole
- d. Baron, M., (2014). Probability & Statistics for Computer Scientists (2nd Ed.), CRC Press

Learning Objectives

01

Solve problems using the counting principle, permutations and combinations.

02

Illustrate random experiment, its sample space and events,

03

State the definitions of probability of an event.

04

Find probabilities that will apply permutations/combinations and the classical definition of probability,

05

Solve problems in Computer Science involving conditional probability, Law of Total Probability, Bayes Rule and Multiplication Rule.



Probability Theory

Branch of Mathematics

- Dealing with chance and uncertainty.

Quantification of Likelihood

- Higher probability means more likely.

Quantification of Uncertainty

- Uncertainty is an objective reality that requires decision-making.



Real World Example

Spam Filtering:

- Have you ever wondered how your email inbox knows which emails are junk? It uses a clever application of probability called Bayes' Rule.
- Spam filters analyze the words in an incoming email and calculate the probability that it's spam. For instance, an email containing words like "winner," "free," "prize," or "cash" has a higher probability of being spam. The filter learns over time by analyzing emails you mark as spam. It constantly updates the probabilities to become smarter at protecting your inbox.
- This is a direct application of conditional probability—calculating the probability of an email being spam, given that it contains certain words.



Real World Example

A/B Testing in Web and App Development:

- Tech companies like Netflix, Spotify, and Amazon need to know which new features or designs their users will like best. To do this, they use a method called A/B testing, which is a real-world probability experiment.
- Here's how it works:
 - They create two versions of a webpage or app feature (Version A and Version B).
 - They randomly show Version A to one group of users and Version B to another.
 - They then measure which version performs better—for example, which version gets more clicks, sign-ups, or purchases.
- Probability theory is used to determine if the difference in performance is statistically significant or if it just happened by chance. This helps companies make data-driven decisions instead of just guessing what users want.



Real World Example

Cryptography and Network Security:

- Randomness and unpredictability are at the heart of keeping digital information secure.
- Probability theory is essential for creating strong cryptographic systems.
- When you create an account online, the system generates complex encryption keys to protect your password. These keys are created using randomized algorithms.
- The security of the system relies on the fact that the probability of a hacker guessing this long, random key is incredibly small—so small that it would take a computer millions of years to guess it through brute force.
- This is a direct application of the counting principles: a larger number of possible keys means a lower probability of a random guess being correct.



Real World Example

Network Communication Protocols:

- When you stream a video or browse a website, data is sent over the internet in small pieces called "packets." Due to network congestion or hardware issues, there's a certain probability that some of these packets might get lost or corrupted along the way.
- Protocols like TCP (Transmission Control Protocol) use probability to handle this uncertainty. The protocol acknowledges the packets that have been received successfully. If a packet isn't acknowledged within a certain time, the protocol assumes it was lost and re-sends it, ensuring that you receive the complete and correct file in the end.



Real World Example

Machine Learning and AI (Computational Statistics):

- Probability is the backbone of modern machine learning and artificial intelligence. Many AI models work by calculating the most likely outcome based on the data they've been trained on.
- Predictive Text: When your phone suggests the next word as you type, it's using a probability model to predict the most likely word to follow your current phrase.
- Recommendation Engines: When Netflix or YouTube suggests what to watch next, the algorithm is calculating the probability that you will enjoy a particular show or video based on your viewing history and what similar users have watched.
- Computer Vision: In self-driving cars, AI systems use probability to identify objects. For instance, the system might determine there is a 95% probability that an object is a pedestrian and a 5% probability it's a shadow, helping the car make a safe decision.





Counting Principles

Learning Objectives

01

Identify different principles of counting.

02

Solve word problems involving principles of counting.



Focuses on efficient ways of counting outcomes, especially for large numbers.

Definition:

- (Multiplication Principle). Suppose an experiment E_1 has n_1 possible outcomes, E_2 has n_2 , up until E_k has n_k . Then the composite experiment $E_1E_2 \dots E_k$ will have $n_1 \times n_2 \times \dots \times n_k$ outcomes.

Calculating RGB Colors:

- In computer vision, computer graphics and web design, colors are often represented using the RGB (Red, Green, Blue) color model. Each color is defined by a combination of red, green, and blue light intensity. In the standard 24-bit color system, each of these three channels can have an integer value from 0 to 255.
- How many unique colors can be represented using the standard RGB model?



Solution:

- By definition of Multiplication Principle. Suppose an experiment E1 has n_1 possible outcomes, E2 has n_2 , up until E_k has n_k . Then the composite experiment $E_1E_2 \dots E_k$ will have $n_1 \times n_2 \times \dots \times n_k$ outcomes.

Calculating RGB Colors:

- Choose the Red value: There are 256 possible values for the red channel (from 0 to 255).
- Choose the Green value: There are 256 possible values for the green channel (from 0 to 255).
- Choose the Blue value: There are 256 possible values for the blue channel (from 0 to 255)
- To find the total number of possible color combinations, we multiply the number of choices for each step:
- Total Colors = (Choices for Red) $\times$ (Choices for Green) $\times$ (Choices for Blue)
- Total Colors = $256 \times 256 \times 256$
- Total Colors = 16,777,216
- Therefore, there are 16,777,216 unique colors that can be represented in the standard RGB system. This is why you often see display or monitor specifications mentioning support for "16.7 million colors."



Calculating IPv4 Addresses:

- Every device connected to the internet, like your laptop, phone, or a web server, needs a unique identifier. In the most common system, this is called an IP address. The original version, known as IPv4 (Internet Protocol version 4), uses a set of four numbers separated by dots, like 192.168.1.1.
- Each of these four numbers is called an octet, and it can have any integer value from 0 to 255.
- How many unique IPv4 addresses are theoretically possible?

Solution:

- Multiplication Principle. We can determine the total number of possible addresses by considering the creation of an address as a four-step process.
- Choose the first octet: There are 256 possible values (0-255).
- Choose the second octet: There are 256 possible values (0-255).
- Choose the third octet: There are 256 possible values (0-255).
- Choose the fourth octet: There are 256 possible values (0-255).
- To find the total number of unique combinations, we multiply the number of possibilities for each octet:
- Total Addresses = (Choices for 1st octet) $\times$ (Choices for 2nd) $\times$ (Choices for 3rd) $\times$ (Choices for 4th)
- Total Addresses = $256 \times 256 \times 256 \times 256$
- Total Addresses = 4,294,967,296
- So, there are theoretically almost 4.3 billion unique IPv4 addresses. This simple calculation explains a major event in computing history: as the internet grew, we began to run out of these addresses, which led to the development of a new system, IPv6, to provide trillions more.





Factorials, Permutations and Combinations

A **factorial** is the product of all positive integers up to a number n . It's a way to calculate the total number of ways to arrange a set of distinct items.

Definition:

$$n! = n \cdot (n - 1) \cdot (n - 2) \cdot (n - 3) \cdot 3 \cdot 2 \cdot 1$$

Working Examples

$$4! = 4 \cdot 3 \cdot 2 \cdot 1$$

$$5! = 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1$$

Real World Example

Scenario Problem: Brute-Forcing a Phone's Lock Pattern

Imagine a simple Android lock screen that uses 4 specific, distinct points (A, B, C, D) that must all be connected in some sequence. A hacker wants to write a script to try every single possible path. How many unique patterns does the script have to try?

Solution:

This is a problem of arranging 4 distinct items (the points).

1. For the first point in the pattern, there are 4 choices.
2. For the second, there are 3 remaining choices.
3. For the third, there are 2 choices left.
4. For the last point, there is only 1 choice.

The total number of unique patterns is: $4! = 4 \cdot 3 \cdot 2 \cdot 1 = 24$

The script would need to try 24 unique combinations to guarantee it unlocks the phone. This concept is central to understanding the complexity of brute-force attacks.



A **permutation** is an ordered arrangement of items. We use it when we need to find the number of ways to choose and arrange r items from a set of n items.

Take aways:

- Key Idea: The order of selection is important.
- No Replacement: Once an item is chosen, it cannot be chosen again.

Formula:

$${}_nP_r = \frac{n!}{(n - r)!}$$

Real World Example

Scenario Problem: Assigning tasks to CPU cores

A multi-core processor has 8 cores available. An operating system needs to schedule 3 distinct, high-priority processes (Process A: video rendering, Process B: data encryption, Process C: system backup). The order of assignment matters because assigning the video task to the fastest core (Core 1) is different from assigning the encryption task to it.

How many different ways can these 3 processes be assigned to the 8 cores?

Solution:

Here, we are arranging $r = 3$ processes among $n = 8$ cores, and the order matters.

The permutation formula then is:

$${}_8P_3 = \frac{8!}{(8-3)!} = \frac{8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} = 8 \cdot 7 \cdot 6 = 336$$

There are 336 different ways the operating system can assign these three specific processes to its 8 cores.



A **combination** is a selection of items where the order of selection is not important

Take aways:

Key Idea: You are just choosing a group; the arrangement within the group doesn't count as a new outcome

Formula:

$${}_nC_r = \binom{n}{r} = \frac{n!}{r!(n-r)!}$$



Real World Example

Scenario Problem: Forming a Code Review Team

After you graduate, you can either work as a technical manager (TM) or a project manager (PM). For PM's you need to form a code review team to review the output the works of the TM's group. There are 10 developers on the project, and you need to select a team of 4 to review a critical piece of new code. How many different teams of 4 can you form?

Solution:

- This is a combination problem because the order in which the developers are selected does not matter. A team of Dev A, Dev B, Dev C, and Dev D is the same team as Dev D, Dev C, Dev B, and Dev A.
- We are choosing $r = 4$ developers from $n = 10$.

$${}_{10}P_4 = \binom{10}{4} = \frac{10!}{4!(10-4)!} = \frac{10!}{4!6!} = \frac{10 \cdot 9 \cdot 8 \cdot 7}{4 \cdot 3 \cdot 2 \cdot 1}$$

There are 210 different possible code review teams that can be formed.



A **combination with replacement** is a selection of items where order doesn't matter, and you can choose the same item multiple times.

Take aways:

- Key Idea: You are choosing r items from n types of items, and the supply of each type is unlimited.
- Analogy: Choosing 3 scoops of ice cream from 5 flavors. You can have three scoops of chocolate

Formula:

$$\binom{r + n - 1}{r} = \frac{(r + n - 1)!}{r!(n - 1)!}$$

Real World Example

Scenario Problem: Configuring a Server's Storage

You are configuring a new server and need to choose 4 hard drives for a storage array. The vendor offers 3 types of drives: a fast SSD, a general-purpose HDD, and a large archive HDD. How many different configurations of 4 drives are possible?

Solution:

- This is a combination with replacement because the order you choose the drives doesn't matter, and you can select the same type of drive multiple times (e.g., you can choose 4 SSDs).
- We are choosing $r = 4$ drives from $n = 3$ types.

$$\binom{4 + 3 - 1}{4} = \frac{(4 + 3 - 1)!}{4!(3 - 1)!} = \frac{6!}{4!2!} = \frac{6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{(4 \cdot 3 \cdot 2 \cdot 1)(2 \cdot 1)} = \frac{6 \cdot 5}{2 \cdot 1} = 15$$

There are 15 different storage configurations possible for the server.

Real World Example

Scenario:

- A start-up fintech rolled out a mobile lending product which requires its new users to create a 4-character PIN. The system has specific rules for creating this PIN to ensure a basic level of security.
- The rules are as follow:
 - The PIN must be exactly 4 characters long.
 - The first character must be an uppercase letter from A to D.
 - The remaining characters should be a digit from 0 to 9.

Problem:

1. What is the total number of possible PINs?
2. Now, suppose the system is updated to be more secure, and the three digits in the PIN are not allowed to be repeated (e.g., C582 is valid, but C551 is not). How many unique PINs can be created under this new rule?
3. Imagine a hacker wants to gain access to a student's account by "brute-forcing" the PIN—that is, by guessing every possible combination. Assume the PIN was created under the original rule where digits can be repeated. What is the probability that the hacker guesses the correct PIN on their very first try?



Real World Example

1. What is the total number of possible PINs?

Solution:

This problem can be solved using the **Multiplication Principle**, which states that if an experiment has multiple parts, the total number of outcomes is the product of the number of outcomes for each part.

We can think of creating a PIN as a 4-step process:

1. Choose the first character: There are 4 possible choices (A, B, C, or D).
2. Choose the second character: There are 10 possible choices for the first digit (0-9).
3. Choose the third character: Since repetition is allowed, there are still 10 choices for the second digit (0-9).
4. Choose the fourth character: Similarly, there are 10 choices for the third digit (0-9).

Applying the Multiplication Principle:

1. Total PINs = (Choices for 1st character) x (Choices for 2nd) x (Choices for 3rd) x (Choices for 4th)
2. Total PINs = $4 \times 10 \times 10 \times 10 = 4,000$

So, there are 4,000 possible unique PINs if digits can be repeated.



Real World Example

1. Now, suppose the system is updated to be more secure, and the three digits in the PIN are not allowed to be repeated (e.g., C582 is valid, but C551 is not). How many unique PINs can be created under this new rule?

Solution:

This is a problem where order matters and we are choosing without replacement, which is a **Permutation**.

1. Choose the first character: There are still 4 choices for the uppercase letter.
2. Choose the second character (first digit): There are 10 choices.
3. Choose the third character (second digit): Since one digit has been used and cannot be repeated, there are only 9 remaining choices.
4. Choose the fourth character (third digit): Since two digits have been used, there are now only 8 remaining choices.

Using the **Multiplication Principle** again: Total PINs = $4 \times 10 \times 9 \times 8 = 2,880$

Alternatively, we can see that choosing 3 unique digits from 10 is a permutation, calculated as: ${}_{10}P_3 = \frac{10!}{(10-3)!} = 720$
We then multiply this result by the 4 choices for the first letter:

$$\text{Total PINs} = 4 \times 720 = 2,880$$

Under the no-repetition rule, there are 2,880 possible unique PINs.



Real World Example

1. Imagine a hacker wants to gain access to a student's account by "brute-forcing" the PIN—that is, by guessing every possible combination. Assume the PIN was created under the original rule where digits can be repeated. What is the probability that the hacker guesses the correct PIN on their very first try?

Solution:

The probability of an event is the ratio of the number of expected outcomes to the total number of possible outcomes in the sample space

$$P(Event) = \frac{\text{Number of expected outcome}}{\text{Total number of possible outcomes}}$$

- The total number of possible outcome is the total number of possible PINs. From Part 1, we know the total number of possible PINs is 4,000.
- The number of expected of outcome is when the hacker is "guessing the single correct PIN in the first try" There is only one correct PIN for the account. Therefore, the number of favorable outcomes is 1.

$$P(\text{Correct guess on first try}) = \frac{1}{4000} = 0.00025 = 0.025\%$$



**THANK YOU FOR
LISTENING**

